

Lost in Instruction: Evaluating Causal Sensitivity and Paraphrase Fragility in Vision-Language-Action Models for Robotic Manipulation

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Abstract

Vision-Language-Action (VLA) models adapt pre-trained VLMs to robotic control. However, static benchmarks obscure whether success stems from language comprehension or visual shortcutting. We present a systematic evaluation of SmolVLA-500M, OpenVLA-7B, and OpenVLA-OFT-7.5B under strict scene-fixed causal constraints. We show that: (1) VLAs exhibit asymmetric causal reliance (dropping to 7.1% TSR on object noun swaps, but retaining 97.5% on verb swaps); (2) Verb cross-attention decays exponentially across transformer depth (12.14x Noun Dominance AAR); (3) Paraphrases severely degrade performance; (4) Failure rollouts exhibit immediate kinematic trajectory divergence at $t_{div} \leq 2$; and (5) Instruction-Contrastive Action Guidance (ICAG, $\alpha=0.5$) rescues paraphrase TSR by up to +23.8 pp without re-training.

1. Introduction

Language-conditioned imitation learning has transformed robotic manipulation by enabling generalist policies to translate natural language prompts into continuous motor control. By coupling large transformer backbones with tokenized action heads, Vision-Language-Action (VLA) models leverage Internet-scale pre-training.

Despite impressive benchmark success, evaluating VLAs under fixed prompt templates creates a fundamental ambiguity: **Does the policy causally parse the semantic structure of the language instruction, or does it utilize the language embedding as a weak contextual trigger while relying on visual shortcuts?**

2. Methodology & Metrics

Scene-Fixed Constraint: $S_0(I_{orig}) = S_0(I_{pert})$ matching initial simulator state hashes exactly.

Change Severity Score (CSS): $CSS = 1 - (TSR_{pert} / TSR_{orig})$.

Attention Allocation Ratio (AAR): $AAR(l) = A_{noun}(l) / A_{verb}(l)$.

Instruction-Contrastive Action Guidance (ICAG): $L_{guided} = L(a|O, I) + \alpha * (L(a|O, I) - L(a|O, I_{blank}))$.

3. Empirical Results

RQ1 Causal Sensitivity: Object noun swaps (`wrong_object`) cause universal performance collapse to 7.1% TSR across all models. Swapping verbs (`wrong_action`) retains up to 97.5% TSR.

RQ2 Paraphrase Fragility: SmolVLA-500M collapses to 4.69% paraphrase TSR (PRIDE 2.7). OpenVLA-OFT maintains 74.22% paraphrase TSR (PRIDE 65.8).

RQ3 Kinematic Divergence: Destructive prompts diverge immediately at $t_{div} \leq 2$ steps ($e(t) > 0.05m$).

RQ4 Attention Mechanics: Verb cross-attention decays across depth, driving AAR to 12.14x in late layers.

RQ5 ICAG Mitigation: ICAG guidance ($\alpha=0.5$) boosts paraphrase TSR by up to +23.8 pp.

RQ6 Horizon Impact: Long-horizon tasks (LIBERO-10) compound language vulnerability (OFT compositional TSR drops to 32.0%).

4. Conclusion

This work establishes that VLAs exhibit asymmetric language reliance (strict noun binding vs verb blindness), proves that verb attention decays across depth (AAR=12.14x), and demonstrates that training-free ICAG logit guidance ($\alpha=0.5$) substantially recovers paraphrase robustness.